



SUJET STATAPP 300 - 2026-2027



Sujet proposé par : Lomo Pedro - pedro.lomo@eui.eu
Organisme : European University Institute
Catégorie : Macroéconomie, Economie Monétaire
Sujet :
The Effects of Geopolitical Risk on Innovation: Evidence from International Patent Data

Description du sujet :

Innovation is a key driver of productivity and long-run economic growth. Over recent decades, innovative activity has also become increasingly international, with firms and researchers relying on cross-border collaboration, international knowledge flows, and globally connected technological networks. At the same time, rising geopolitical tensions, trade conflicts, and strategic competition are reshaping international economic relations. These developments may therefore affect not only the overall level of innovation, but also where innovation takes place, in which technological sectors, and how knowledge flows across countries.

The aim of the proposed study is to investigate how geopolitical risk affects innovative activity using international patent data. The project will first examine whether changes in geopolitical risk are associated with changes in patenting activity across countries and over time. It will then study whether these effects differ across technological sectors, with the objective of understanding whether geopolitical tensions lead to a reallocation of innovative activity towards or away from particular technologies.

A second dimension of the project will focus on the international organization of innovation. Using patent citation data and, where available, information on inventors and co-invention, students will investigate whether geopolitical tensions affect cross-border knowledge flows and international research collaboration. This will allow the project to assess whether geopolitical fragmentation changes not only the quantity of innovation, but also its geographical and technological composition.

Finally, the project may provide a more detailed application to France. In particular, students may examine how French patenting has evolved across technological sectors, which areas of the French innovation system appear more exposed to geopolitical tensions, and whether changes in geopolitical risk are associated with shifts in the sectoral or international composition of French innovative activity.

Données du sujet :

? OECD patent and patent citation databases (other alternatives also available), which can be used to construct measures of innovative activity and international knowledge flows;

? Measures of geopolitical risk;

? Macroeconomic and country and industry-level control variables from international databases such as the World Bank, OECD, or IMF.

The precise level of aggregation, sample period, and set of variables will be determined according to data availability and the empirical questions developed during the project.

Modèles envisagés du sujet :

Students will be expected to conduct descriptive analysis and apply appropriate statistical and econometric methods to panel data. Depending on the development of the project, the analysis may include panel fixed-effects regressions, dynamic specifications, heterogeneous effects across countries and sectors, count models (e.g. Poisson) for patent counts, network analyses, gravity models.

Students will mainly work with R or Python.

Bibliographie du sujet :

Aiyar, S., et al. (2023). "Geeoeconomic Fragmentation and the Future of Multilateralism." IMF Staff Discussion Note SDN/2023/001.

Autor, D., Dorn, D., Hanson, G. H., Pisano, G., & Shu, P. (2020). "Foreign Competition and Domestic Innovation: Evidence from US Patents." American Economic Review: Insights, 2(3), 357?374.

Caldara, D. and Iacoviello, M. (2022). ?Measuring Geopolitical Risk.? American Economic Review, 112(4), 1194?1225.

Griliches, Z. (1990). "Patent Statistics as Economic Indicators: A Survey." Journal of Economic Literature, 28(4), 1661?1707.

Jaffe, A. B., Trajtenberg, M. and Henderson, R. (1993). ?Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations.? Quarterly Journal of Economics, 108(3), 577?598.

Notation du sujet :**Commentaires :**